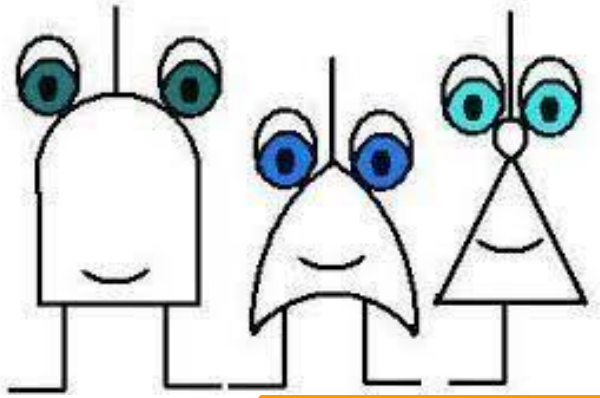
LOGIC GATES and TRUTH TABLES

Lecture 3

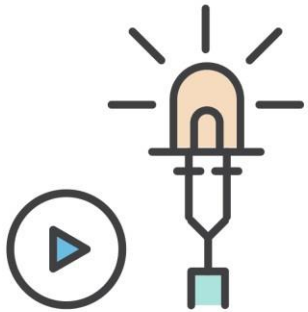


1

WHAT ARE LOGIC GATES?

WHAT ARE LOGIC GATES?

- Logic gates are the basic building blocks of any digital system.
- Using these logic gates, you can build digital logic circuits for different applications
- They are implemented inside the ICs. (Integrated circuits)



WHAT ARE LOGIC GATES?

- Each logic gate performs a specific logical operations and associated with a logic expression.
- Each logic gate has a symbol and a truth table of its own.
- Truth table is a way of expressing the logical operation in detail.

2

3 BASIC LOGIC GATES

3 BASIC LOGIC GATES

1. NOT
2. AND
3. OR

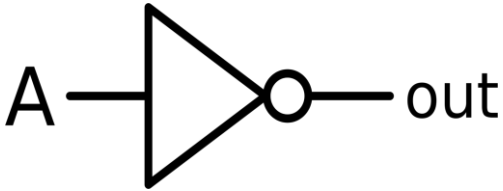
1. NOT (Inverter) Gate

- Logical Operation: Performs logical negation on its input.
- Logic Expression: $\bar{A}$

$Y = \bar{A}$ or

$Y = A'$

Symbol



Truth Table

Input	Output
0	**1**
1	**0**

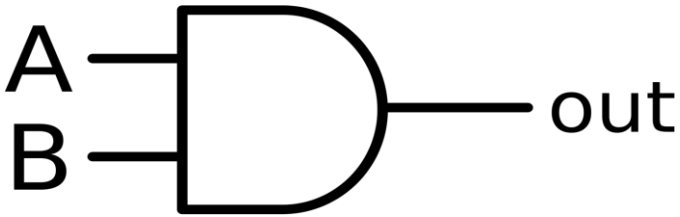
0 - Low Voltage

1 - High Voltage

2. AND Gate

- Logical Operation: Implements logical conjunction
- Logic Expression: Y = A.B

Symbol



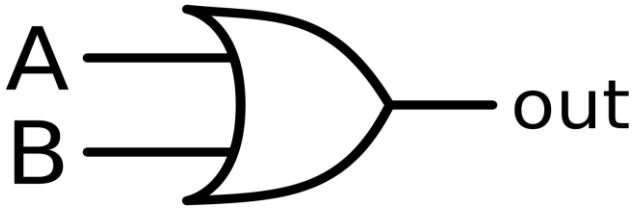
Truth Table

A	B	Output
0	**0**	**0**
0	**1**	**0**
1	**0**	**0**
1	**1**	**1**

3. OR GATE

- Logical Operation: Implements logical inclusive
- Logic Expression: Y = A + B

Symbol



Truth Table

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

3

DERIVED LOGIC GATES

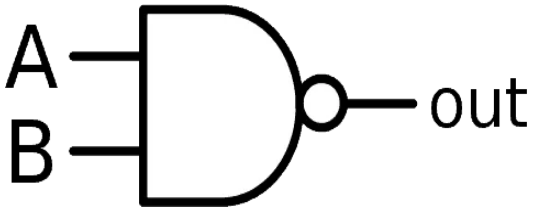
DERIVED LOGIC GATES

1. NAND
2. NOR
3. XOR
4. XNOR

4. NAND Gate

- Logical Operation: AND with a NOT
- Logic Expression: $Y = \overline{A.B}$

Symbol



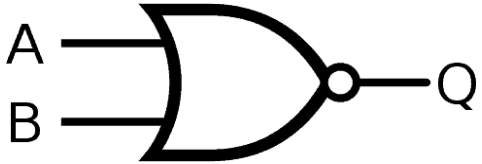
Truth Table

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR Gate

- Logical Operation: OR with a NOT
- Logic Expression: $Y = \overline{A + B}$

Symbol



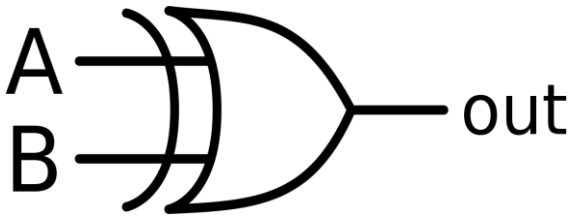
Truth Table

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

6. XOR Gate

- Logical Operation: Exclusive OR Gate
- Logic Expression: $Y = A \oplus B$

Symbol



Truth Table

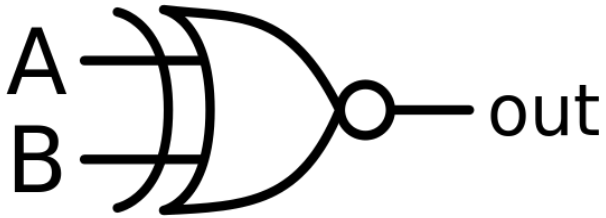
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Output will be high when the inputs are different from each other

7. XNOR Gate

- Logical Operation: Exclusive NOR Gate
- Logic Expre ssion: $Y = \overline{A \oplus B}$

Symbol



Truth Table

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Output will be high when the inputs are same

More about Logic Gates

- NOT is a unary operator, always has one input
- AND, OR, NAND, NOR gates usually have two inputs but there can be more than two inputs for these gates.
- Then they are called Multiple Input Logic Gates.
- Remember XOR and XNOR are always two input gates

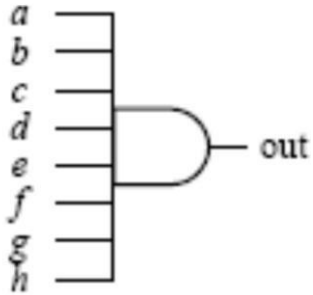
More about Logic Gates



→ Four input AND gate - All four inputs must be high in order to make the output high



→ Three input OR gate - At least one input out of three has to be high in order to make the output high



→ Eight input AND gate - All the eight inputs must be high for
output to be high

Summary

Will watch a video now!





THANKS!

Any questions?